

Key: Two Systems, One Map

84-355 Democracy's Data | Census geography | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure, as of the rebuild. The brief (`census-geography-brief.Rmd`) and the lab (`census-geography.Rmd`) run the *same numbered steps* — the brief in prose with D3 figures, the lab with the code exposed. **Students should read the brief first.** Step numbers below refer to both.

Teach this EARLY. It is the foundation five later labs assume and none of them supplies — precincts against blocks, HOLC areas against tracts, EAVS jurisdictions, §203 thresholds, BISG. Run before those and each of them gets a sentence shorter and a great deal clearer. Run after, and it works as the session where everything that went wrong is explained at once.

It is also the shortest lab in the course. Six small tables, three maps of one county, no modelling. Use the time it saves on the discussion.

The maps are all Houston County, Georgia — the same county the precinct-geography lab uses, so students meet the ground twice. Say that out loud; the second encounter is worth much more if they know it is the second.

Every number below was run against the data files.

The session in one sentence

American geography is two systems — one where the identifier contains the hierarchy and every join is exact, and one where it does not and every join is an assumption — and almost every data problem in this course lives on the seam.

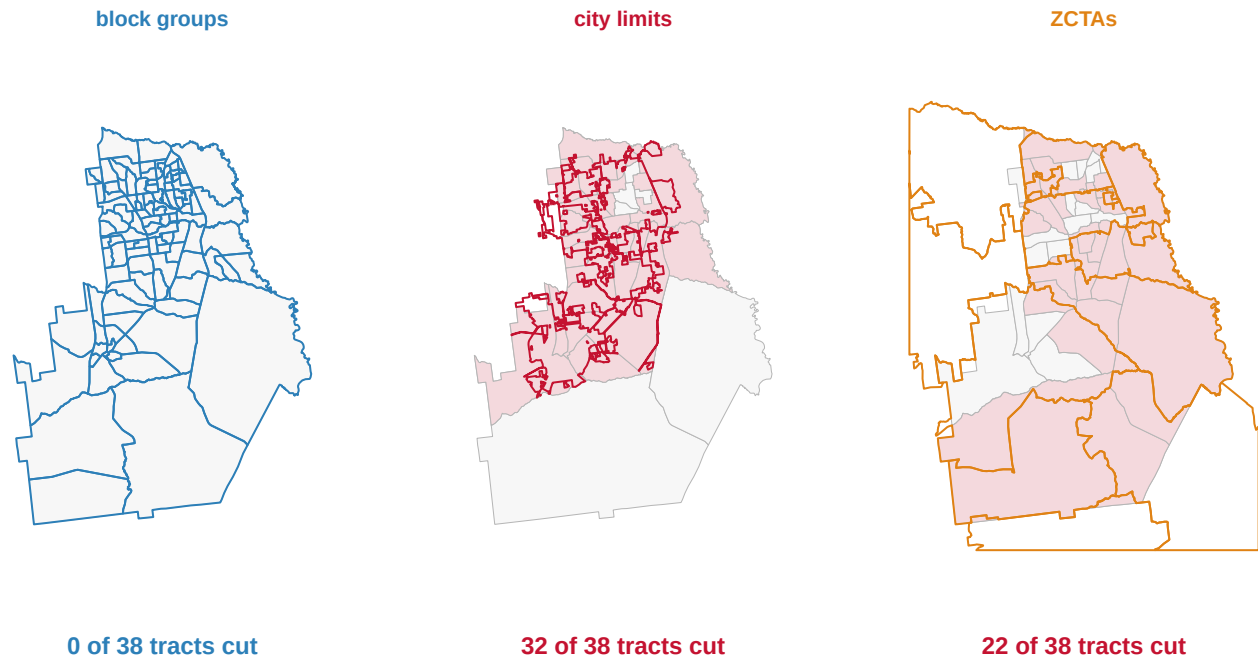
The findings

```
c(counties = lv$count[lv$geography == "county"],
  tracts   = lv$count[lv$geography == "census tract"],
  zctas    = lv$count[lv$geography == "ZCTA (ZIP approximation)"],
  tracts_nesting_cleanly = n$n_pass[1],
  zcta_file_has_a_state   = n$result[3])
```

```
#>           counties           tracts           zctas
#>           "3222"           "85396"         "33791"
#> tracts_nesting_cleanly zcta_file_has_a_state
#>           "85396"           "False"
```

- **85,396 of 85,396 tracts** match a real county by ID prefix. Zero orphans.
- **The ZCTA file has no state column and no state in the ID.**
- A median county is **604 sq mi**; a median tract **1.74**; a median ZCTA **34.89** — twenty tracts' worth.

And the same two claims drawn, on Houston County, Georgia (3,269 blocks, 108 block groups, 38 tracts):



- A block group boundary cuts **0** of the county's 38 tracts. A city limit cuts **32** of them, and a ZCTA boundary cuts **22**.
- Warner Robins is in **2 counties**, and 5 of the county's 12 ZCTAs are in more than one county. That is the identifier's missing county component, on the ground.

The three things to get said

1. The GEOID is the join. Students think of an ID as an arbitrary key. Write the block identifier from Step 1 on the board — 131530211233013 — and take it apart: state, county, tract, block group, block. **Once they see that the first five digits *are* the county, the nesting spine stops being a fact to memorise and becomes something they can read off any file.** The five-panel figure in Step 1 is the same five pieces as places; put them side by side.

2. Nesting is by construction, not by luck. The Census Bureau draws small units inside large ones on purpose, so tracts nest perfectly and always will. Places, ZIP codes and school districts were drawn by other people, for other reasons, at other times. **Nothing forces them to line up, and they don't.** The three panels in Step 7 are the whole argument in one row of pictures: the geography that nests cuts zero tracts, the two that do not cut 32 and 22.

3. The ZIP code is the one they will actually use. Every student will one day be handed data “by ZIP.” Make sure they leave knowing it is a **mail route, not an area** — a set of delivery stops the Postal Service changes when routing changes. The Census’s ZCTA is an approximation of something that is not a shape. **Point at the file: no state column, because some ZIPs are in two states.**

The line for the board: if the ID contains the hierarchy, the join is free. Otherwise somebody has to decide.

The move that makes this session land

Ask the room where their ZIP code is. They took the census replication survey on the first day and typed one in. Then show them Step 8.

It reframes the whole thing from an abstraction about identifiers into something they personally got wrong two months ago, and it costs ninety seconds.

Walking the steps

Steps	What they do	Min	Watch for
1-3	One GEOID; the five maps it names; the gazetteer; how many of each	9	The identifier is built by concatenation. That claim is testable, and the zoom sequence is the same claim as geography.
4	Test it: all 85,396 tracts match a county by prefix; the county at four scales	9	Zero orphans. Nesting is not documentation, it is arithmetic — and no line is drawn twice.
5-7	The two tests that could not be run; what each ID encodes; the three overlays counted	14	The pivot. An empty result column because the file has no such field, then 0 / 32 / 22 tracts cut.
8	ZIP codes	8	Five digits encoding nothing — and the first-day survey asked for one.
9-10	Why this bit four other labs; what you can say	10	Precincts, HOLC, EAVS jurisdictions, Cook County. One cause.

Step 5 is the pivot. A test that *cannot be run* is a stronger result than one that fails: the Bureau declined to record the relationship, because the relationship does not exist.

If you are short on time, protect the pivot and the closing step. The numbers are what make the argument concrete; the sequence is the argument.

Option A — Build the hierarchy

Verified output

```
lv[, c("geography", "nests_into", "id_length")]
```

```
#>           geography nests_into id_length
#> 1             county      state         5
#> 2      census tract      county        11
#> 3             place state only         7
#> 4  ZCTA (ZIP approximation)  NOTHING         5
#> 5      county subdivision      county        10
#> 6 elementary school district state only         7
#> 7 unified school district state only         7
```

The tree: **state** → **county** → { **census tract** → **block group** → **block**, **county subdivision** }.
Outside it: places, ZCTAs, both kinds of school district.

3/4: Draws the tree correctly and puts the crosscutting units to one side.

4/4: Says what the outsiders have in common — **they were all drawn by somebody other than the Census Bureau, for a purpose other than counting people**. Municipalities incorporate, school districts are created by states, ZIP codes serve mail. The Bureau records them; it does not design them.

4/4, best: Notices that “nests into state only” is doing sneaky work. A place ID contains a state, so places *do* nest into states — but that is nearly useless, because the join anybody wants is to a county or a tract. **A hierarchy that only reaches the top level is not a hierarchy you can use.**

Option B — How wrong is a ZIP?

Verified output

```
z <- lv$median_sq_mi[lv$geography == "ZCTA (ZIP approximation)"]
t <- lv$median_sq_mi[lv$geography == "census tract"]
c(median_zcta = z, median_tract = t, ratio = round(z / t, 1))
```

```
#> median_zcta median_tract      ratio
#>      34.89       1.74      20.10
```

About twenty tracts to a ZCTA by area.

3/4: Reports the ratio and says ZIP-level statistics average over a lot.

4/4: Makes it concrete — a ZIP-level median income is one number for an area that might contain both the richest and poorest tracts in a city, and the averaging is not random, it is *spatial*. Neighbourhoods differ block to block; that is the entire finding of the redlining lab.

4/4, best: Notices the comparison is by **area**, not population, and that these are very different in a city — a dense urban ZCTA might contain many more than twenty tracts while a rural one contains a fraction of one. **The twenty-times figure is itself an average that hides its own variance**, which is the lesson eating its own tail, appropriately.

Option C — The school district trap

No verified output beyond the counts; grade the reasoning.

1,952 elementary districts, 10,893 unified.

3/4: Works out that a student in an elementary district must attend a secondary district for high school, so the two overlap.

4/4: Draws the consequence for data — **the same child is inside two districts at once**, so “students per district” double-counts unless you know which system you are in, and a map of “school districts” is either incomplete or overlapping depending on which files you loaded.

4/4, best: Connects it back to the main lesson: this is not a quirk of schools, it is what happens whenever two authorities draw boundaries over the same ground for different purposes. **Congressional districts, precincts, water boards, fire districts — the voter file in Session 6 had twenty-one such columns for exactly this reason.**

Option D — Read the middle panel

Verified output

```
sp[, c("overlay", "role", "units", "tracts_total", "tracts_split")]
```

```
#>               overlay      role units tracts_total tracts_split
#> 1           block group on the spine    108           38           0
#> 2 census place (city limits) crosscutting      4           38          32
#> 3      ZCTA (ZIP approximation) crosscutting    12           38          22
```

The question asks what you must assume to report a tract-level statistic for a city whose limits cut 32 of 38 tracts.

3/4: Says you would have to split the cut tracts somehow, and names area or population weighting.

4/4: States the assumption in its strong form — **that the people inside the city limits look like the people outside them, within the same tract** — and notices that this is exactly the assumption the tract was drawn to avoid making, since tracts are meant to be internally similar and a city limit is not.

4/4, best: Names a case where it fails hard and says why. Annexation is selective: cities annex subdivisions and shopping centres and leave trailer parks and farmland outside, so the line between incorporated and

unincorporated land inside one tract is frequently a line between different kinds of household. **The error is not random, it is correlated with exactly the thing you are measuring.** Anyone who gets there has understood the lab.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is explaining *why* the nesting units nest — by construction — rather than treating it as a happy accident.
- **A 2** is “you should use tracts instead of ZIP codes” with no account of why, or of what to do when the data only comes by ZIP.
- **Reward anyone who reconsiders an earlier lab** in light of this one. That is the whole point of running it.

Anticipated questions

- “*Can I convert ZIP to tract?*” — Crosswalks exist (HUD publishes one) and they are **allocations, not translations** — each ZIP maps to several tracts with weights. Fine if you say so; wrong if you pretend it is a lookup.
- “*Why does anyone publish by ZIP then?*” — Because it is the only geography respondents know. That is a real answer, not a dodge, and it is why the first- day survey asked for one.
- “*Do blocks nest into precincts?*” — **No, and that is Session 14’s whole problem.** Good moment to forward-reference it.
- “*What about congressional districts?*” — They nest into states and nothing else, and they are redrawn every decade, so they do not even nest into their own past. The gazetteer’s CD file was unavailable when this was built.
- “*Is a ‘place’ the same as a city?*” — Roughly, plus census-designated places, which are unincorporated areas the Bureau treats as places for counting. A place can exist for statistics and not for government. The county mapped here has three incorporated cities and one CDP, **Robins AFB**, which is the base: a place for counting, not a municipality.
- “*Why this county?*” — Because the precinct-geography lab uses it too, so the ground is already familiar by the second time they see it, and because it is small enough that all 3,269 of its blocks fit on a page. Nothing about the argument depends on the choice; the numbers do.
- “*Is 32 of 38 typical?*” — **Unknown from this lab**, and worth saying so. It is one county with a heavily annexed city in it. The direction of the result is structural; the size of it is local.
- “*Why do blocks nest into everything?*” — Because the Bureau draws them last, against every boundary it knows about, which is why a block is the atom the crosswalks in later labs are built on. **Blocks nest into places and ZCTAs; tracts do not.** That is the sentence to leave them with.

Connections

This lab is a prerequisite for five others, and each should point back:

- **Session 6 (voter files)** — 21 of 53 columns are overlapping districts.
- **Session 11 (EAVS)** — “jurisdiction” varies by state.
- **Session 14 (precincts)** — precincts against blocks, the same seam.

- **Redlining** — 1937 areas against 2020 tracts, joined by centroid.
- **Session 7 (BISG)** — works *because* blocks nest. The easy case, and worth saying so, since students otherwise never notice when a join went right.